

Comprehensive viva questions (Benny)

- 1 Explain how to improve volumetric efficiency of an engine.
- 2 Explain how volumetric efficiency related to emissions.
- 3 What are the main pollutant from diesel engine and gasoline engine?
- 4 Explain how to measure the NO_x.
- 5 What is the unit of combustion (degree crank angle or sec)
- 6 What is the difference between combustion in turbine and gasoline engine.
- 7 What are the stages in diesel engine and gasoline engine?
- 8 What is knocking in SI engine and CI engine?
- 9 How to measure knock in a diesel engine?
- 10 Explain cetain number and octain number?
- 11 Write Nevier Stock equation in vector form.
- 12 Write it for X momentum.
- 13 Whether it is for compressible flow or not.
- 14 What is the change in this equation if consider turbulence?
- 15 What type of equation it is.?
- 16 How to find out it hyperbolic, parabolic or elliptical?
- 17 What is Raynolds Number?
- 18 Write equation for pressure drop in pipe flow.
- 19 Write Darcy's equation for frictional head loss.
- 20 What is Burnollis equation. For unsteady flow.
- 21 What are the factors affecting frictional head.
- 22 If we using different metals with same roughness what will be the effect on friction loss.
- 23 What method is using in Star CD?
- 24 Explain finite element and finite volume. Explain.
- 25 Explain different combustion models?
- 26 Write continuity equation.
- 27 What type of equation it is.
- 28 What methods are adopted for increasing the air intake to the engine?
- 29 How to control flow of fuel or air in to engine by electronically.
- 30 Explain how the grid moving in the star cd .
- 31 What type of cells used in the model?
- 32 Given some values of x (say 2,3,4,6,7,8) and corresponding values of y (say 3,5,7,6.8,9,7). Write a mathematical equation for finding out any value of Y corresponding to X. (say x=2.5).

Comprehensive viva voce Questions:

1. What is cycle efficiency?
2. What is its effect on Mechanical and Combustion efficiency?
3. What is overall efficiency expression?
4. State first law of thermodynamics.
5. What is a closed system and open system?
6. What is internal energy of a system?
7. What is a point function and a path function?
8. Is the extended expansion concept can be applied to constant pressure cycle?
9. Show the valve timing on pV diagram.
10. What is the maximum efficiency that is expected from extended expansion engine?
11. What are the modifications that are to be done to convert a conventional engine to extended expansion engine?
12. What is the maximum cylinder pressure and exhaust manifold pressure? (Numerical values)
13. What are the cycles, which work and have maximum efficiency as Carnot cycle?
14. What is Mollier diagram? Show a refrigeration cycle on it.
15. What is the difference between Pdv & Vdp ?
16. What are Maxwell's equations and state their significance?
17. Even though IC engine is a open system why its working is explained by a closed cycle? (pV diagram)

H.R.Purushothama.
(ME03D037)

Comprehensive Exam Questions (Anand)

1. Define swirl and explain with illustration
2. What is meant by swirl ratio?
3. Explain how you can measure swirl in engine?
4. What is effect of engine speed on swirl?
5. Explain IC engine processes
6. What is flammable mixture
7. Explain flammability limits. Give the flammability limits for SI engine.
8. Explain different lubricating systems in IC Engines
9. Zeldovich mechanism and write rate equation for formation of NO
10. What is the driving potential in mass transfer?
11. Governing laws in mass transfer.
12. What is convection and give types of convection
13. Explain significance of Grashoff number in natural convection.
14. Give any example where both natural and forced convection exists
15. Explain flow through the pipe.
16. How do you measure pressure loss in the pipe?
17. What is meant by friction factor?
18. What are the boundary conditions applied for a pipe flow
19. What is fully developed flow
20. Types of flow, give some examples for external flows
21. Explain flow over a sphere.
22. What do you mean by vortex flow, free vortex and forced vortex.
23. What are the different types of equations (elliptical, parabolic, hyperbolic)
24. What is the nature of Navier – Stokes equation
25. What are the parameters going to measure in flow field analysis for your geometry?

COMPREHENSIVE EXAM QUESTIONS (jyothish)

From turbulence

1. What is prandtl's mixing length?
2. What is the expression for the above and how will you derive it?
3. What are the assumptions that we make near laminar sub layer?
4. What is RANS?
5. Give an equation how will you do time averaging of that? Where you will get the extra terms.

(given equation is $\partial u / \partial t + u \partial u / \partial x = \nu \partial^2 u / \partial y^2$)

From measurements

6. What are the different flow measuring instruments?

7. Draw the venturimeter? Write the expression for finding the flow rate?

From thermodynamics and combustion

8. What is entropy?
9. Two glasses one with water, and another with water and ice, they are maintaining at the same temperature. What will happen to the entropy.
10. What is fugacity?

From simulation

11. What is simulation?
12. What are the different models?
13. What do you mean by phenomenological model.?
14. What is SMD?
15. Explain how spray is forming from the droplet? (given in our professor's book)

About optimization

16. How will you solve an optimization problem? (refer non linear programming)
About objective functions,
About constraints

From aerodynamics

17. What is singularity?
18. Where this singularity comes in a flow?
19. How will you simulate a symmetrical aerofoil using these elementary flows?
20. What will happen if you superimpose a vortex flow over a source.

COMPREHENSIVE EXAM QUESTIONS (james)

1. Define Combustion?
2. What are the auto ignition temperatures of gasoline and Diesel?
3. What will happen to the fuel when it is heated to its self ignition temperature?
4. Define directional derivative?
5. Define a parabolic partial differential equation?
6. Write a discretized equation of a simple unsteady heat conduction?
7. Define vorticity?
8. Write the Navier stokes equation?
9. How many stress components are there in a Navier stokes equation?

10. Draw the boundary layer profile of velocity in a pipe line?
11. Explain the significance of Reynolds number in a pipeline flow? Why the Reynolds number changes say from 2000 in a laminar region to more than 5000 in a turbulent region?
12. Draw the flow through a pipe line and show the boundary conditions to be applied to find its exit velocity?
13. Write a general chemical equation of burning of an organic chemical substance which has the formula like: C_xH_y ? Write down the Air fuel ratio in terms of x and y?
14. How will you measure the HC emission from an engine?
15. Explain the principle behind the measurement of NO_x emission from an engine?
16. What are cool flames? Draw the curve?
17. What is open cycle?
18. Draw the open cycle gas turbine?
19. Draw the closed cycle turbine?
20. What is the difference between the actual Gas turbine and theoretical cycle?
21. Write down the expression for the work done during the compression process in a Brayton cycle? List out the assumptions made in the expression